

UPSC Civil Services Preliminary Examination - CSAT Paper II

DETAILED SOLUTIONS / ANSWER KEY | Time: 2 Hours | Maximum Marks: 200

Quick Key

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Detailed solutions

Q1. Answer: A | Reading Comprehension

Passage: A district digitised land records, but dispute resolution did not improve until maps, mutation registers and field verification were reconciled. Technology reduced search time; it could not decide contested title by itself. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which option best states the central idea?

- A. Digitisation improves access but cannot replace legal and field reconciliation. - CORRECT: Correct. The central idea joins the intervention's benefit to the condition that makes it effective.
- B. The intervention should be abandoned in every setting. - INCORRECT: Incorrect: The intervention should be abandoned in every setting. does not follow the stated data. The central idea joins the intervention's benefit to the condition that makes it effective.
- C. The passage proves that implementation conditions never matter. - INCORRECT: Incorrect: The passage proves that implementation conditions never matter. is an intermediate, sign or operation error. The central idea joins the intervention's benefit to the condition that makes it effective.
- D. The passage recommends maximising only the most visible input. - INCORRECT: Incorrect: The passage recommends maximising only the most visible input. ignores a condition or uses the wrong base. The central idea joins the intervention's benefit to the condition that makes it effective.

Method / explanation: The central idea joins the intervention's benefit to the condition that makes it effective.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q2. Answer: B | Reading Comprehension

Passage: A district digitised land records, but dispute resolution did not improve until maps, mutation registers and field verification were reconciled. Technology reduced search time; it could not decide contested title by itself. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which inference is best supported?

- A. Every stakeholder responds identically in all circumstances. - INCORRECT: Incorrect: Every stakeholder responds identically in all circumstances. does not follow the stated data. The inference is limited to what the passage's comparison supports.
- B. Faster retrieval and accurate adjudication are different outcomes. - CORRECT: Correct. The inference is limited to what the passage's comparison supports.
- C. The opposite outcome must occur whenever one condition changes. - INCORRECT: Incorrect: The opposite outcome must occur whenever one condition changes. is an intermediate, sign or operation error. The inference is limited to what the passage's comparison supports.

D. A policy not discussed in the passage is legally compulsory. - INCORRECT: Incorrect: A policy not discussed in the passage is legally compulsory. ignores a condition or uses the wrong base. The inference is limited to what the passage's comparison supports.

Method / explanation: The inference is limited to what the passage's comparison supports.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q3. Answer: C | Reading Comprehension

Passage: A district digitised land records, but dispute resolution did not improve until maps, mutation registers and field verification were reconciled. Technology reduced search time; it could not decide contested title by itself. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which assumption is necessary for the reasoning?

A. The intervention has no cost under any condition. - INCORRECT: Incorrect: The intervention has no cost under any condition. does not follow the stated data. The necessary assumption connects the observed mechanism to the stated conclusion.

B. No alternative explanation can ever exist. - INCORRECT: Incorrect: No alternative explanation can ever exist. is an intermediate, sign or operation error. The necessary assumption connects the observed mechanism to the stated conclusion.

C. Citizens contesting title require records that refer to the same parcel. - CORRECT: Correct. The necessary assumption connects the observed mechanism to the stated conclusion.

D. All participants possess identical preferences and resources. - INCORRECT: Incorrect: All participants possess identical preferences and resources. ignores a condition or uses the wrong base. The necessary assumption connects the observed mechanism to the stated conclusion.

Method / explanation: The necessary assumption connects the observed mechanism to the stated conclusion.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q4. Answer: D | Reading Comprehension

Passage: A district digitised land records, but dispute resolution did not improve until maps, mutation registers and field verification were reconciled. Technology reduced search time; it could not decide contested title by itself. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. The author's tone is best described as:

A. unreservedly celebratory - INCORRECT: Incorrect: unreservedly celebratory does not follow the stated data. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

B. hostile and dismissive - INCORRECT: Incorrect: hostile and dismissive is an intermediate, sign or operation error. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

C. unrelated and indifferent - INCORRECT: Incorrect: unrelated and indifferent ignores a condition or uses the wrong base. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

D. qualified and analytical - CORRECT: Correct. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

Method / explanation: The passage recognises both achievement and limitation, so an absolute tone is unsupported.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q5. Answer: A | Reading Comprehension

Passage: A self-help group began collective procurement of seed. Costs fell, yet the poorest members benefited only after instalment payments were allowed. Scale created efficiency; payment design determined inclusion. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which option best states the central idea?

A. Collective buying needs inclusive payment design to share its gains. - CORRECT: Correct. The central idea joins the intervention's benefit to the condition that makes it effective.

B. The intervention should be abandoned in every setting. - INCORRECT: Incorrect: The intervention should be abandoned in every setting. does not follow the stated data. The central idea joins the intervention's benefit to the condition that makes it effective.

C. The passage proves that implementation conditions never matter. - INCORRECT: Incorrect: The passage proves that implementation conditions never matter. is an intermediate, sign or operation error. The central idea joins the intervention's benefit to the condition that makes it effective.

D. The passage recommends maximising only the most visible input. - INCORRECT: Incorrect: The passage recommends maximising only the most visible input. ignores a condition or uses the wrong base. The central idea joins the intervention's benefit to the condition that makes it effective.

Method / explanation: The central idea joins the intervention's benefit to the condition that makes it effective.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q6. Answer: B | Reading Comprehension

Passage: A self-help group began collective procurement of seed. Costs fell, yet the poorest members benefited only after instalment payments were allowed. Scale created efficiency; payment design determined inclusion. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which inference is best supported?

A. Every stakeholder responds identically in all circumstances. - INCORRECT: Incorrect: Every stakeholder responds identically in all circumstances. does not follow the stated data. The inference is limited to what the passage's comparison supports.

B. Lower unit cost alone need not make an input affordable. - CORRECT: Correct. The inference is limited to what the passage's comparison supports.

C. The opposite outcome must occur whenever one condition changes. - INCORRECT: Incorrect: The opposite outcome must occur whenever one condition changes. is an intermediate, sign or operation error. The inference is limited to what the passage's comparison supports.

D. A policy not discussed in the passage is legally compulsory. - INCORRECT: Incorrect: A policy not discussed in the passage is legally compulsory. ignores a condition or uses the wrong base. The inference is limited to what the passage's comparison supports.

Method / explanation: The inference is limited to what the passage's comparison supports.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q7. Answer: C | Reading Comprehension

Passage: A self-help group began collective procurement of seed. Costs fell, yet the poorest members benefited only after instalment payments were allowed. Scale created efficiency; payment design determined inclusion. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which assumption is necessary for the reasoning?

A. The intervention has no cost under any condition. - INCORRECT: Incorrect: The intervention has no cost under any condition. does not follow the stated data. The necessary assumption connects the observed mechanism to the stated conclusion.

B. No alternative explanation can ever exist. - INCORRECT: Incorrect: No alternative explanation can ever exist. is an intermediate, sign or operation error. The necessary assumption connects the observed mechanism to the stated conclusion.

C. Some poorer members faced a liquidity constraint. - CORRECT: Correct. The necessary assumption connects the observed mechanism to the stated conclusion.

D. All participants possess identical preferences and resources. - INCORRECT: Incorrect: All participants possess identical preferences and resources. ignores a condition or uses the wrong base. The necessary assumption connects the observed mechanism to the stated conclusion.

Method / explanation: The necessary assumption connects the observed mechanism to the stated conclusion.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q8. Answer: D | Reading Comprehension

Passage: A self-help group began collective procurement of seed. Costs fell, yet the poorest members benefited only after instalment payments were allowed. Scale created efficiency; payment design determined inclusion. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. The author's tone is best described as:

A. unreservedly celebratory - INCORRECT: Incorrect: unreservedly celebratory does not follow the stated data. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

B. hostile and dismissive - INCORRECT: Incorrect: hostile and dismissive is an intermediate, sign or operation error. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

C. unrelated and indifferent - INCORRECT: Incorrect: unrelated and indifferent ignores a condition or uses the wrong base. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

D. balanced - CORRECT: Correct. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

Method / explanation: The passage recognises both achievement and limitation, so an absolute tone is unsupported.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q9. Answer: A | Reading Comprehension

Passage: A city planted roadside trees but survival remained low where contracts rewarded planting rather than maintenance. When payments were linked to three-year survival, contractors changed species choice and watering schedules. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which option best states the central idea?

A. Outcome-linked contracts can improve urban tree survival. - CORRECT: Correct. The central idea joins the intervention's benefit to the condition that makes it effective.

B. The intervention should be abandoned in every setting. - INCORRECT: Incorrect: The intervention should be abandoned in every setting. does not follow the stated data. The central idea joins the intervention's benefit to the condition that makes it effective.

C. The passage proves that implementation conditions never matter. - INCORRECT: Incorrect: The passage proves that implementation conditions never matter. is an intermediate, sign or operation error. The central idea joins the intervention's benefit to the condition that makes it effective.

D. The passage recommends maximising only the most visible input. - INCORRECT: Incorrect: The passage recommends maximising only the most visible input. ignores a condition or uses the wrong base. The central idea joins the intervention's benefit to the condition that makes it effective.

Method / explanation: The central idea joins the intervention's benefit to the condition that makes it effective.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q10. Answer: B | Reading Comprehension

Passage: A city planted roadside trees but survival remained low where contracts rewarded planting rather than maintenance. When payments were linked to three-year survival, contractors changed species choice and watering schedules. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which inference is best supported?

A. Every stakeholder responds identically in all circumstances. - INCORRECT: Incorrect: Every stakeholder responds identically in all circumstances. does not follow the stated data. The inference is limited to what the passage's comparison supports.

B. Incentives influenced maintenance and species choice. - CORRECT: Correct. The inference is limited to what the passage's comparison supports.

C. The opposite outcome must occur whenever one condition changes. - INCORRECT: Incorrect: The opposite outcome must occur whenever one condition changes. is an intermediate, sign or operation error. The inference is limited to what the passage's comparison supports.

D. A policy not discussed in the passage is legally compulsory. - INCORRECT: Incorrect: A policy not discussed in the passage is legally compulsory. ignores a condition or uses the wrong base. The inference is limited to what the passage's comparison supports.

Method / explanation: The inference is limited to what the passage's comparison supports.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q11. Answer: C | Reading Comprehension

Passage: A city planted roadside trees but survival remained low where contracts rewarded planting rather than maintenance. When payments were linked to three-year survival, contractors changed species choice and watering schedules. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which assumption is necessary for the reasoning?

A. The intervention has no cost under any condition. - INCORRECT: Incorrect: The intervention has no cost under any condition. does not follow the stated data. The necessary assumption connects the observed mechanism to the stated conclusion.

B. No alternative explanation can ever exist. - INCORRECT: Incorrect: No alternative explanation can ever exist. is an intermediate, sign or operation error. The necessary assumption connects the observed mechanism to the stated conclusion.

C. Survival can be measured reliably enough to affect payment. - CORRECT: Correct. The necessary assumption connects the observed mechanism to the stated conclusion.

D. All participants possess identical preferences and resources. - INCORRECT: Incorrect: All participants possess identical preferences and resources. ignores a condition or uses the wrong base. The necessary assumption connects the observed mechanism to the stated conclusion.

Method / explanation: The necessary assumption connects the observed mechanism to the stated conclusion.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q12. Answer: D | Reading Comprehension

Passage: A city planted roadside trees but survival remained low where contracts rewarded planting rather than maintenance. When payments were linked to three-year survival, contractors changed species choice and watering schedules. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. The author's tone is best described as:

A. unreservedly celebratory - INCORRECT: Incorrect: unreservedly celebratory does not follow the stated data. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

B. hostile and dismissive - INCORRECT: Incorrect: hostile and dismissive is an intermediate, sign or operation error. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

C. unrelated and indifferent - INCORRECT: Incorrect: unrelated and indifferent ignores a condition or uses the wrong base. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

D. evidence-based - CORRECT: Correct. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

Method / explanation: The passage recognises both achievement and limitation, so an absolute tone is unsupported.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q13. Answer: A | Reading Comprehension

Passage: A school served breakfast before morning classes. Attendance rose, but learning gains were largest where teachers also used short diagnostic exercises. Nutrition improved readiness; instruction converted readiness into learning. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which option best states the central idea?

A. Nutrition and responsive teaching work as complements. - CORRECT: Correct. The central idea joins the intervention's benefit to the condition that makes it effective.

B. The intervention should be abandoned in every setting. - INCORRECT: Incorrect: The intervention should be abandoned in every setting. does not follow the stated data. The central idea joins the intervention's benefit to the condition that makes it effective.

C. The passage proves that implementation conditions never matter. - INCORRECT: Incorrect: The passage proves that implementation conditions never matter. is an intermediate, sign or operation error. The central idea joins the intervention's benefit to the condition that makes it effective.

D. The passage recommends maximising only the most visible input. - INCORRECT: Incorrect: The passage recommends maximising only the most visible input. ignores a condition or uses the wrong base. The central idea joins the intervention's benefit to the condition that makes it effective.

Method / explanation: The central idea joins the intervention's benefit to the condition that makes it effective.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q14. Answer: B | Reading Comprehension

Passage: A school served breakfast before morning classes. Attendance rose, but learning gains were largest where teachers also used short diagnostic exercises. Nutrition improved readiness; instruction converted readiness into learning. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which inference is best supported?

A. Every stakeholder responds identically in all circumstances. - INCORRECT: Incorrect: Every stakeholder responds identically in all circumstances. does not follow the stated data. The inference is limited to what the passage's comparison supports.

B. Attendance gains do not automatically equal learning gains. - CORRECT: Correct. The inference is limited to what the passage's comparison supports.

C. The opposite outcome must occur whenever one condition changes. - INCORRECT: Incorrect: The opposite outcome must occur whenever one condition changes. is an intermediate, sign or operation error. The inference is limited to what the passage's comparison supports.

D. A policy not discussed in the passage is legally compulsory. - INCORRECT: Incorrect: A policy not discussed in the passage is legally compulsory. ignores a condition or uses the wrong base. The inference is limited to what the passage's comparison supports.

Method / explanation: The inference is limited to what the passage's comparison supports.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q15. Answer: C | Reading Comprehension

Passage: A school served breakfast before morning classes. Attendance rose, but learning gains were largest where teachers also used short diagnostic exercises. Nutrition improved readiness; instruction converted readiness into learning. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which assumption is necessary for the reasoning?

A. The intervention has no cost under any condition. - INCORRECT: Incorrect: The intervention has no cost under any condition. does not follow the stated data. The necessary assumption connects the observed mechanism to the stated conclusion.

B. No alternative explanation can ever exist. - INCORRECT: Incorrect: No alternative explanation can ever exist. is an intermediate, sign or operation error. The necessary assumption connects the observed mechanism to the stated conclusion.

C. Diagnostic exercises helped teachers adapt instruction. - CORRECT: Correct. The necessary assumption connects the observed mechanism to the stated conclusion.

D. All participants possess identical preferences and resources. - INCORRECT: Incorrect: All participants possess identical preferences and resources. ignores a condition or uses the wrong base. The necessary assumption connects the observed mechanism to the stated conclusion.

Method / explanation: The necessary assumption connects the observed mechanism to the stated conclusion.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q16. Answer: D | Reading Comprehension

Passage: A school served breakfast before morning classes. Attendance rose, but learning gains were largest where teachers also used short diagnostic exercises. Nutrition improved readiness; instruction converted readiness into learning. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. The author's tone is best described as:

A. unreservedly celebratory - INCORRECT: Incorrect: unreservedly celebratory does not follow the stated data. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

B. hostile and dismissive - INCORRECT: Incorrect: hostile and dismissive is an intermediate, sign or operation error. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

C. unrelated and indifferent - INCORRECT: Incorrect: unrelated and indifferent ignores a condition or uses the wrong base. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

D. measured - CORRECT: Correct. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

Method / explanation: The passage recognises both achievement and limitation, so an absolute tone is unsupported.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q17. Answer: A | Reading Comprehension

Passage: A coastal village received cyclone warnings in time, yet evacuation lagged until local volunteers translated the message into landmarks and arranged transport for older residents. Forecast accuracy mattered, but last-mile trust and capability completed the warning chain. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which option best states the central idea?

A. Early warning succeeds only when trusted last-mile action is possible. - CORRECT: Correct. The central idea joins the intervention's benefit to the condition that makes it effective.

B. The intervention should be abandoned in every setting. - INCORRECT: Incorrect: The intervention should be abandoned in every setting. does not follow the stated data. The central idea joins the intervention's benefit to the condition that makes it effective.

C. The passage proves that implementation conditions never matter. - INCORRECT: Incorrect: The passage proves that implementation conditions never matter. is an intermediate, sign or operation error. The central idea joins the intervention's benefit to the condition that makes it effective.

D. The passage recommends maximising only the most visible input. - INCORRECT: Incorrect: The passage recommends maximising only the most visible input. ignores a condition or uses the wrong base. The central idea joins the intervention's benefit to the condition that makes it effective.

Method / explanation: The central idea joins the intervention's benefit to the condition that makes it effective.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q18. Answer: B | Reading Comprehension

Passage: A coastal village received cyclone warnings in time, yet evacuation lagged until local volunteers translated the message into landmarks and arranged transport for older residents. Forecast accuracy mattered, but last-mile trust and capability completed the warning chain. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which inference is best supported?

A. Every stakeholder responds identically in all circumstances. - INCORRECT: Incorrect: Every stakeholder responds identically in all circumstances. does not follow the stated data. The inference is limited to what the passage's comparison supports.

B. Technical forecasts can fail to produce evacuation without local capability. - CORRECT: Correct. The inference is limited to what the passage's comparison supports.

C. The opposite outcome must occur whenever one condition changes. - INCORRECT: Incorrect: The opposite outcome must occur whenever one condition changes. is an intermediate, sign or operation error. The inference is limited to what the passage's comparison supports.

D. A policy not discussed in the passage is legally compulsory. - INCORRECT: Incorrect: A policy not discussed in the passage is legally compulsory. ignores a condition or uses the wrong base. The inference is limited to what the passage's comparison supports.

Method / explanation: The inference is limited to what the passage's comparison supports.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q19. Answer: C | Reading Comprehension

Passage: A coastal village received cyclone warnings in time, yet evacuation lagged until local volunteers translated the message into landmarks and arranged transport for older residents. Forecast accuracy mattered, but last-mile trust and capability completed the warning chain. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. Which assumption is necessary for the reasoning?

A. The intervention has no cost under any condition. - INCORRECT: Incorrect: The intervention has no cost under any condition. does not follow the stated data. The necessary assumption connects the observed mechanism to the stated conclusion.

B. No alternative explanation can ever exist. - INCORRECT: Incorrect: No alternative explanation can ever exist. is an intermediate, sign or operation error. The necessary assumption connects the observed mechanism to the stated conclusion.

C. Residents understood and trusted locally translated guidance. - CORRECT: Correct. The necessary assumption connects the observed mechanism to the stated conclusion.

D. All participants possess identical preferences and resources. - INCORRECT: Incorrect: All participants possess identical preferences and resources. ignores a condition or uses the wrong base. The necessary assumption connects the observed mechanism to the stated conclusion.

Method / explanation: The necessary assumption connects the observed mechanism to the stated conclusion.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q20. Answer: D | Reading Comprehension

Passage: A coastal village received cyclone warnings in time, yet evacuation lagged until local volunteers translated the message into landmarks and arranged transport for older residents. Forecast accuracy mattered, but last-mile trust and capability completed the warning chain. The episode also warns against reading an aggregate improvement as proof that every group benefited equally. A sound evaluation must distinguish the formal input, the mechanism through which people can use it, and the distribution of gains or burdens when implementation conditions differ. The author's tone is best described as:

- A. unreservedly celebratory - INCORRECT: Incorrect: unreservedly celebratory does not follow the stated data. The passage recognises both achievement and limitation, so an absolute tone is unsupported.
- B. hostile and dismissive - INCORRECT: Incorrect: hostile and dismissive is an intermediate, sign or operation error. The passage recognises both achievement and limitation, so an absolute tone is unsupported.
- C. unrelated and indifferent - INCORRECT: Incorrect: unrelated and indifferent ignores a condition or uses the wrong base. The passage recognises both achievement and limitation, so an absolute tone is unsupported.
- D. cautionary - CORRECT: Correct. The passage recognises both achievement and limitation, so an absolute tone is unsupported.

Method / explanation: The passage recognises both achievement and limitation, so an absolute tone is unsupported.

Trap: Choose the qualified claim; reject universal language not supported by the passage.

Q21. Answer: A | Numeracy

What is the remainder when $154^2 + 3 \times 154 + 5$ is divided by 8?

- A. 7 - CORRECT: Correct. Reduce 154 modulo 8, then evaluate $r^2 + 3r + 5$ modulo 8; the remainder is 7.
- B. 0 - INCORRECT: Incorrect: 0 does not follow the stated data. Reduce 154 modulo 8, then evaluate $r^2 + 3r + 5$ modulo 8; the remainder is 7.
- C. 1 - INCORRECT: Incorrect: 1 is an intermediate, sign or operation error. Reduce 154 modulo 8, then evaluate $r^2 + 3r + 5$ modulo 8; the remainder is 7.
- D. 2 - INCORRECT: Incorrect: 2 ignores a condition or uses the wrong base. Reduce 154 modulo 8, then evaluate $r^2 + 3r + 5$ modulo 8; the remainder is 7.

Method / explanation: Reduce 154 modulo 8, then evaluate $r^2 + 3r + 5$ modulo 8; the remainder is 7.

Trap: Reduce each term modulo the divisor; do not expand the large square unnecessarily.

Q22. Answer: B | Numeracy

Two numbers are obtained by doubling 20 and tripling 33. What is their HCF?

- A. 2 - INCORRECT: Incorrect: 2 does not follow the stated data. The derived numbers are 40 and 99; Euclid's algorithm gives $\gcd(40, 99) = 1$.
- B. 1 - CORRECT: Correct. The derived numbers are 40 and 99; Euclid's algorithm gives $\gcd(40, 99) = 1$.
- C. 3 - INCORRECT: Incorrect: 3 is an intermediate, sign or operation error. The derived numbers are 40 and 99; Euclid's algorithm gives $\gcd(40, 99) = 1$.
- D. 3960 - INCORRECT: Incorrect: 3960 ignores a condition or uses the wrong base. The derived numbers are 40 and 99; Euclid's algorithm gives $\gcd(40, 99) = 1$.

Method / explanation: The derived numbers are 40 and 99; Euclid's algorithm gives $\gcd(40, 99) = 1$.

Trap: Apply the transformation before computing the HCF.

Q23. Answer: C | Numeracy

How many positive divisors does 36 have?

- A. 8 - INCORRECT: Incorrect: 8 does not follow the stated data. $36 = 2^2 \times 3^2$, so divisor count = $(3)(3) = 9$.
- B. 10 - INCORRECT: Incorrect: 10 is an intermediate, sign or operation error. $36 = 2^2 \times 3^2$, so divisor count = $(3)(3) = 9$.
- C. 9 - CORRECT: Correct. $36 = 2^2 \times 3^2$, so divisor count = $(3)(3) = 9$.
- D. 6 - INCORRECT: Incorrect: 6 ignores a condition or uses the wrong base. $36 = 2^2 \times 3^2$, so divisor count = $(3)(3) = 9$.

Method / explanation: $36 = 2^2 \times 3^2$, so divisor count = $(3)(3) = 9$.

Trap: Add one to every prime exponent before multiplying.

Q24. Answer: D | Numeracy

What is the unit digit of $(4^{12}) \times (8^6)$?

A. 5 - INCORRECT: Incorrect: 5 does not follow the stated data. Find each unit-digit cycle separately and multiply the two terminal digits modulo 10; the result is 4.

B. 6 - INCORRECT: Incorrect: 6 is an intermediate, sign or operation error. Find each unit-digit cycle separately and multiply the two terminal digits modulo 10; the result is 4.

C. 7 - INCORRECT: Incorrect: 7 ignores a condition or uses the wrong base. Find each unit-digit cycle separately and multiply the two terminal digits modulo 10; the result is 4.

D. 4 - CORRECT: Correct. Find each unit-digit cycle separately and multiply the two terminal digits modulo 10; the result is 4.

Method / explanation: Find each unit-digit cycle separately and multiply the two terminal digits modulo 10; the result is 4.

Trap: Do not add the unit digits of the factors.

Q25. Answer: A | Numeracy

The arithmetic sequence begins 4, 7, 10. What is the sum of its first 9 terms?

A. 144.0 - CORRECT: Correct. $S_n = n/2[2a + (n-1)d] = 9/2[2 \times 4 + 8 \times 3] = 144.0$.

B. 141.0 - INCORRECT: Incorrect: 141.0 does not follow the stated data. $S_n = n/2[2a + (n-1)d] = 9/2[2 \times 4 + 8 \times 3] = 144.0$.

C. 147.0 - INCORRECT: Incorrect: 147.0 is an intermediate, sign or operation error. $S_n = n/2[2a + (n-1)d] = 9/2[2 \times 4 + 8 \times 3] = 144.0$.

D. 28 - INCORRECT: Incorrect: 28 ignores a condition or uses the wrong base. $S_n = n/2[2a + (n-1)d] = 9/2[2 \times 4 + 8 \times 3] = 144.0$.

Method / explanation: $S_n = n/2[2a + (n-1)d] = 9/2[2 \times 4 + 8 \times 3] = 144.0$.

Trap: The last term is not the sum.

Q26. Answer: B | Numeracy

After multiplying 9987 by 9, what is the sum of the decimal digits of the product?

A. 35 - INCORRECT: Incorrect: 35 does not follow the stated data. $9987 \times 9 = 89883$; adding the digits of 89883 gives 36.

B. 36 - CORRECT: Correct. $9987 \times 9 = 89883$; adding the digits of 89883 gives 36.

C. 37 - INCORRECT: Incorrect: 37 is an intermediate, sign or operation error. $9987 \times 9 = 89883$; adding the digits of 89883 gives 36.

D. 38 - INCORRECT: Incorrect: 38 ignores a condition or uses the wrong base. $9987 \times 9 = 89883$; adding the digits of 89883 gives 36.

Method / explanation: $9987 \times 9 = 89883$; adding the digits of 89883 gives 36.

Trap: The divisibility-by-9 check verifies the digit sum but does not by itself give its unreduced value.

Q27. Answer: C | Numeracy

How many integers from 16 through 122, inclusive, are divisible by 7 but not by 14?

A. 7 - INCORRECT: Incorrect: 7 does not follow the stated data. Count multiples of 7 (15) and subtract multiples of 14 (7); result 8.

B. 9 - INCORRECT: Incorrect: 9 is an intermediate, sign or operation error. Count multiples of 7 (15) and subtract multiples of 14 (7); result 8.

C. 8 - CORRECT: Correct. Count multiples of 7 (15) and subtract multiples of 14 (7); result 8.

D. 10 - INCORRECT: Incorrect: 10 ignores a condition or uses the wrong base. Count multiples of 7 (15) and subtract multiples of 14 (7); result 8.

Method / explanation: Count multiples of 7 (15) and subtract multiples of 14 (7); result 8.

Trap: Multiples of the larger divisor are a subset and must be excluded once.

Q28. Answer: D | Numeracy

The values are 13, 19, 25, 31. If the smallest falls by 1 and the largest rises by 2, what is the new mean?

A. 22.0 - INCORRECT: Incorrect: 22.0 does not follow the stated data. The adjusted values are [12, 19, 25, 33]; their total 89 divided by 4 gives 22.25.

B. 23.25 - INCORRECT: Incorrect: 23.25 is an intermediate, sign or operation error. The adjusted values are [12, 19, 25, 33]; their total 89 divided by 4 gives 22.25.

C. 24.25 - INCORRECT: Incorrect: 24.25 ignores a condition or uses the wrong base. The adjusted values are [12, 19, 25, 33]; their total 89 divided by 4 gives 22.25.

D. 22.25 - CORRECT: Correct. The adjusted values are [12, 19, 25, 33]; their total 89 divided by 4 gives 22.25.

Method / explanation: The adjusted values are [12, 19, 25, 33]; their total 89 divided by 4 gives 22.25.

Trap: Apply both changes to the total before dividing.

Q29. Answer: A | Algebra

If $3x + 2 = 20$, what is x ?

A. 6 - CORRECT: Correct. Subtract 2 and divide by 3: $x = 6$.

B. 5 - INCORRECT: Incorrect: 5 does not follow the stated data. Subtract 2 and divide by 3: $x = 6$.

C. 7 - INCORRECT: Incorrect: 7 is an intermediate, sign or operation error. Subtract 2 and divide by 3: $x = 6$.

D. 18 - INCORRECT: Incorrect: 18 ignores a condition or uses the wrong base. Subtract 2 and divide by 3: $x = 6$.

Method / explanation: Subtract 2 and divide by 3: $x = 6$.

Trap: Undo addition before division.

Q30. Answer: B | Algebra

If $2(x - 1) = 8$, what is x when x is a positive integer and the displayed equality is supplemented by $x + 1 = 6$?

A. 4 - INCORRECT: Incorrect: 4 does not follow the stated data. The second equation fixes x : $x + 1 = 6$, hence $x = 5$.

B. 5 - CORRECT: Correct. The second equation fixes x : $x + 1 = 6$, hence $x = 5$.

C. 6 - INCORRECT: Incorrect: 6 is an intermediate, sign or operation error. The second equation fixes x : $x + 1 = 6$, hence $x = 5$.

D. 10 - INCORRECT: Incorrect: 10 ignores a condition or uses the wrong base. The second equation fixes x : $x + 1 = 6$, hence $x = 5$.

Method / explanation: The second equation fixes x : $x + 1 = 6$, hence $x = 5$.

Trap: An identity alone cannot determine x ; use the independent equation.

Q31. Answer: C | Algebra

The sum of two consecutive integers is 19. What is the smaller integer?

A. 10 - INCORRECT: Incorrect: 10 does not follow the stated data. Let the integers be x and $x+1$. Then $2x+1=19$, so $x=9$.

B. 8 - INCORRECT: Incorrect: 8 is an intermediate, sign or operation error. Let the integers be x and $x+1$. Then $2x+1=19$, so $x=9$.

C. 9 - CORRECT: Correct. Let the integers be x and $x+1$. Then $2x+1=19$, so $x=9$.

D. 19 - INCORRECT: Incorrect: 19 ignores a condition or uses the wrong base. Let the integers be x and $x+1$. Then $2x+1=19$, so $x=9$.

Method / explanation: Let the integers be x and $x+1$. Then $2x+1=19$, so $x=9$.

Trap: The question asks for the smaller integer.

Q32. Answer: D | Algebra

Five years ago A was half of A's age 12 years from now. What is A's present age?

A. 17 - INCORRECT: Incorrect: 17 does not follow the stated data. Let present age be x . Then $x-5 = (x+12)/2$; solving gives $x=22$.

B. 27 - INCORRECT: Incorrect: 27 is an intermediate, sign or operation error. Let present age be x . Then $x-5 = (x+12)/2$; solving gives $x=22$.

C. 11 - INCORRECT: Incorrect: 11 ignores a condition or uses the wrong base. Let present age be x . Then $x-5 = (x+12)/2$; solving gives $x=22$.

D. 22 - CORRECT: Correct. Let present age be x . Then $x-5 = (x+12)/2$; solving gives $x=22$.

Method / explanation: Let present age be x . Then $x-5 = (x+12)/2$; solving gives $x=22$.

Trap: Translate both time references from the present.

Q33. Answer: A | Algebra

What is the positive solution of $x^2 = 49$?

- A. 7 - CORRECT: Correct. $x = \text{plus or minus } 7$; the positive solution is 7.
- B. -7 - INCORRECT: Incorrect: -7 does not follow the stated data. $x = \text{plus or minus } 7$; the positive solution is 7.
- C. 49 - INCORRECT: Incorrect: 49 is an intermediate, sign or operation error. $x = \text{plus or minus } 7$; the positive solution is 7.
- D. 8 - INCORRECT: Incorrect: 8 ignores a condition or uses the wrong base. $x = \text{plus or minus } 7$; the positive solution is 7.
- Method / explanation: $x = \text{plus or minus } 7$; the positive solution is 7.
- Trap: The word positive removes the negative root.

Q34. Answer: B | Algebra

Which value is the least integer satisfying $2x + 1 > 16$?

- A. 7 - INCORRECT: Incorrect: 7 does not follow the stated data. $2x > 15$, so $x > 7.5$; the least integer is 8.
- B. 8 - CORRECT: Correct. $2x > 15$, so $x > 7.5$; the least integer is 8.
- C. 9 - INCORRECT: Incorrect: 9 is an intermediate, sign or operation error. $2x > 15$, so $x > 7.5$; the least integer is 8.
- D. 16 - INCORRECT: Incorrect: 16 ignores a condition or uses the wrong base. $2x > 15$, so $x > 7.5$; the least integer is 8.
- Method / explanation: $2x > 15$, so $x > 7.5$; the least integer is 8.
- Trap: A strict inequality excludes the boundary.

Q35. Answer: C | Percentages

A grant equals 15% of Rs 280. An administrative surcharge of 6% is then applied to the grant. What is the final amount?

- A. 42 - INCORRECT: Incorrect: 42 does not follow the stated data. First grant= $15/100 \times 280 = 42$; then multiply by 106/100 to get 44.52.
- B. 38.52 - INCORRECT: Incorrect: 38.52 is an intermediate, sign or operation error. First grant= $15/100 \times 280 = 42$; then multiply by 106/100 to get 44.52.
- C. 44.52 - CORRECT: Correct. First grant= $15/100 \times 280 = 42$; then multiply by 106/100 to get 44.52.
- D. 58.8 - INCORRECT: Incorrect: 58.8 ignores a condition or uses the wrong base. First grant= $15/100 \times 280 = 42$; then multiply by 106/100 to get 44.52.
- Method / explanation: First grant= $15/100 \times 280 = 42$; then multiply by 106/100 to get 44.52.
- Trap: The surcharge applies to the grant, not to the original amount.

Q36. Answer: D | Percentages

A value of 1000 rises by 9% and then falls by 4%. What is the final value?

- A. 1050 - INCORRECT: Incorrect: 1050 does not follow the stated data. Successive multipliers give $1000 \times 1.09 \times 0.96 = 1046.4$.
- B. 1050.0 - INCORRECT: Incorrect: 1050.0 is an intermediate, sign or operation error. Successive multipliers give $1000 \times 1.09 \times 0.96 = 1046.4$.
- C. 1056.4 - INCORRECT: Incorrect: 1056.4 ignores a condition or uses the wrong base. Successive multipliers give $1000 \times 1.09 \times 0.96 = 1046.4$.
- D. 1046.4 - CORRECT: Correct. Successive multipliers give $1000 \times 1.09 \times 0.96 = 1046.4$.
- Method / explanation: Successive multipliers give $1000 \times 1.09 \times 0.96 = 1046.4$.
- Trap: Successive percentages are not simply netted.

Q37. Answer: A | Ratios

Rs 480 is divided in the ratio 3:3. What is the first share?

- A. 240.0 - CORRECT: Correct. First share = $480 \times 3/(3+3) = 240.0$.
- B. 237.0 - INCORRECT: Incorrect: 237.0 does not follow the stated data. First share = $480 \times 3/(3+3) = 240.0$.
- C. 243.0 - INCORRECT: Incorrect: 243.0 is an intermediate, sign or operation error. First share = $480 \times 3/(3+3) = 240.0$.
- D. 80.0 - INCORRECT: Incorrect: 80.0 ignores a condition or uses the wrong base. First share = $480 \times 3/(3+3) = 240.0$.
- Method / explanation: First share = $480 \times 3/(3+3) = 240.0$.
- Trap: Use ratio parts, not either raw ratio number.

Q38. Answer: B | Averages

Groups of 21 and 31 have averages 41 and 61. What is their combined average?

- A. 51.0 - INCORRECT: Incorrect: 51.0 does not follow the stated data. Weighted mean = $(21 \times 41 + 31 \times 61) / (52) = 52.92$.
- B. 52.92 - CORRECT: Correct. Weighted mean = $(21 \times 41 + 31 \times 61) / (52) = 52.92$.
- C. 51.92 - INCORRECT: Incorrect: 51.92 is an intermediate, sign or operation error. Weighted mean = $(21 \times 41 + 31 \times 61) / (52) = 52.92$.
- D. 53.92 - INCORRECT: Incorrect: 53.92 ignores a condition or uses the wrong base. Weighted mean = $(21 \times 41 + 31 \times 61) / (52) = 52.92$.

Method / explanation: Weighted mean = $(21 \times 41 + 31 \times 61) / (52) = 52.92$.

Trap: Do not average group averages without weights.

Q39. Answer: C | Percentages

An article costs Rs 550, is marked for a 16% profit, and then receives a 5% rebate on that marked selling price. What is the realised price?

- A. 638.0 - INCORRECT: Incorrect: 638.0 does not follow the stated data. Marked selling price=638.0; realised price= $638.0 \times 95 / 100 = 606.1$.
- B. 610.5 - INCORRECT: Incorrect: 610.5 is an intermediate, sign or operation error. Marked selling price=638.0; realised price= $638.0 \times 95 / 100 = 606.1$.
- C. 606.1 - CORRECT: Correct. Marked selling price=638.0; realised price= $638.0 \times 95 / 100 = 606.1$.
- D. 611.1 - INCORRECT: Incorrect: 611.1 ignores a condition or uses the wrong base. Marked selling price=638.0; realised price= $638.0 \times 95 / 100 = 606.1$.

Method / explanation: Marked selling price=638.0; realised price= $638.0 \times 95 / 100 = 606.1$.

Trap: Profit and rebate use different successive bases.

Q40. Answer: D | Percentages

A marked price of Rs 900 is discounted by 11%. What is the sale price?

- A. 889 - INCORRECT: Incorrect: 889 does not follow the stated data. Sale price = $900 \times 89 / 100 = 801.0$.
- B. 999.0 - INCORRECT: Incorrect: 999.0 is an intermediate, sign or operation error. Sale price = $900 \times 89 / 100 = 801.0$.
- C. 790.0 - INCORRECT: Incorrect: 790.0 ignores a condition or uses the wrong base. Sale price = $900 \times 89 / 100 = 801.0$.
- D. 801.0 - CORRECT: Correct. Sale price = $900 \times 89 / 100 = 801.0$.

Method / explanation: Sale price = $900 \times 89 / 100 = 801.0$.

Trap: A percentage discount is not a rupee subtraction.

Q41. Answer: A | Ratios

A mixture contains milk and water in the ratio 4:5. In 36 litres, how many litres are milk?

- A. 16.0 - CORRECT: Correct. Milk = $36 \times 4 / (4+5) = 16.0$.
- B. 15.0 - INCORRECT: Incorrect: 15.0 does not follow the stated data. Milk = $36 \times 4 / (4+5) = 16.0$.
- C. 4.0 - INCORRECT: Incorrect: 4.0 is an intermediate, sign or operation error. Milk = $36 \times 4 / (4+5) = 16.0$.
- D. 17.0 - INCORRECT: Incorrect: 17.0 ignores a condition or uses the wrong base. Milk = $36 \times 4 / (4+5) = 16.0$.

Method / explanation: Milk = $36 \times 4 / (4+5) = 16.0$.

Trap: The first term of the ratio denotes milk.

Q42. Answer: B | Averages

The average of 6 values is 41. After adding 53, what is the new average?

- A. 41 - INCORRECT: Incorrect: 41 does not follow the stated data. Old total 246; new average = $(246+53) / 7 = 42.71$.
- B. 42.71 - CORRECT: Correct. Old total 246; new average = $(246+53) / 7 = 42.71$.
- C. 47.0 - INCORRECT: Incorrect: 47.0 is an intermediate, sign or operation error. Old total 246; new average = $(246+53) / 7 = 42.71$.

D. 43.71 - INCORRECT: Incorrect: 43.71 ignores a condition or uses the wrong base. Old total 246; new average = $(246+53)/7 = 42.71$.

Method / explanation: Old total 246; new average = $(246+53)/7 = 42.71$.

Trap: Recover the old total before adding a value.

Q43. Answer: C | Time and Work

A can finish a task in 9 days and B in 13 days. How many days will they take together?

A. 22 - INCORRECT: Incorrect: 22 does not follow the stated data. Combined rate = $1/9+1/13$; time = $117/(22) = 5.32$ days.

B. 11.0 - INCORRECT: Incorrect: 11.0 is an intermediate, sign or operation error. Combined rate = $1/9+1/13$; time = $117/(22) = 5.32$ days.

C. 5.32 - CORRECT: Correct. Combined rate = $1/9+1/13$; time = $117/(22) = 5.32$ days.

D. 4 - INCORRECT: Incorrect: 4 ignores a condition or uses the wrong base. Combined rate = $1/9+1/13$; time = $117/(22) = 5.32$ days.

Method / explanation: Combined rate = $1/9+1/13$; time = $117/(22) = 5.32$ days.

Trap: Add rates, not times.

Q44. Answer: D | Time and Work

13 workers finish a job in 15 days at equal efficiency. How many days will 19 workers take?

A. 15 - INCORRECT: Incorrect: 15 does not follow the stated data. Worker-days stay constant: $13 \times 15 = 19x$, so $d=10.26$.

B. 21.92 - INCORRECT: Incorrect: 21.92 is an intermediate, sign or operation error. Worker-days stay constant: $13 \times 15 = 19x$, so $d=10.26$.

C. 11.26 - INCORRECT: Incorrect: 11.26 ignores a condition or uses the wrong base. Worker-days stay constant: $13 \times 15 = 19x$, so $d=10.26$.

D. 10.26 - CORRECT: Correct. Worker-days stay constant: $13 \times 15 = 19x$, so $d=10.26$.

Method / explanation: Worker-days stay constant: $13 \times 15 = 19x$, so $d=10.26$.

Trap: For fixed work, workers and days vary inversely.

Q45. Answer: A | Time and Work

Two pipes fill a tank in 7 and 9 hours. If opened together, how long do they take?

A. 3.94 - CORRECT: Correct. Rates add: $1/7+1/9$; reciprocal gives 3.94 hours.

B. 16 - INCORRECT: Incorrect: 16 does not follow the stated data. Rates add: $1/7+1/9$; reciprocal gives 3.94 hours.

C. 8.0 - INCORRECT: Incorrect: 8.0 is an intermediate, sign or operation error. Rates add: $1/7+1/9$; reciprocal gives 3.94 hours.

D. 4.94 - INCORRECT: Incorrect: 4.94 ignores a condition or uses the wrong base. Rates add: $1/7+1/9$; reciprocal gives 3.94 hours.

Method / explanation: Rates add: $1/7+1/9$; reciprocal gives 3.94 hours.

Trap: Both are inlet pipes.

Q46. Answer: B | Time and Work

An inlet fills a tank in 6 hours while a leak empties it in 21 hours. How long to fill together?

A. 4.67 - INCORRECT: Incorrect: 4.67 does not follow the stated data. Net rate = $1/6-1/21$; time = 8.40 hours.

B. 8.4 - CORRECT: Correct. Net rate = $1/6-1/21$; time = 8.40 hours.

C. 15 - INCORRECT: Incorrect: 15 is an intermediate, sign or operation error. Net rate = $1/6-1/21$; time = 8.40 hours.

D. 6 - INCORRECT: Incorrect: 6 ignores a condition or uses the wrong base. Net rate = $1/6-1/21$; time = 8.40 hours.

Method / explanation: Net rate = $1/6-1/21$; time = 8.40 hours.

Trap: An outlet rate must be subtracted.

Q47. Answer: C | Time and Work

A is 3 times as efficient as B. If B alone takes 30 days, how many days does A take?

A. 30 - INCORRECT: Incorrect: 30 does not follow the stated data. Time is inverse to efficiency: $30/3=10$ days.

B. 3 - INCORRECT: Incorrect: 3 is an intermediate, sign or operation error. Time is inverse to efficiency: $30/3=10$ days.

C. 10 - CORRECT: Correct. Time is inverse to efficiency: $30/3=10$ days.

D. 13 - INCORRECT: Incorrect: 13 ignores a condition or uses the wrong base. Time is inverse to efficiency: $30/3=10$ days.

Method / explanation: Time is inverse to efficiency: $30/3=10$ days.

Trap: Greater efficiency means less time.

Q48. Answer: D | Time and Work

A team completes 0.37 of a job. What fraction remains, rounded to two decimals?

- A. 0.37 - INCORRECT: Incorrect: 0.37 does not follow the stated data. Remaining work = $1 - 0.37 = 0.63$.
- B. 1.37 - INCORRECT: Incorrect: 1.37 is an intermediate, sign or operation error. Remaining work = $1 - 0.37 = 0.63$.
- C. 0.73 - INCORRECT: Incorrect: 0.73 ignores a condition or uses the wrong base. Remaining work = $1 - 0.37 = 0.63$.
- D. 0.63 - CORRECT: Correct. Remaining work = $1 - 0.37 = 0.63$.

Method / explanation: Remaining work = $1 - 0.37 = 0.63$.

Trap: Completed and remaining fractions sum to one.

Q49. Answer: A | Speed and Distance

A vehicle's total journey time is 4 hours, including a halt of 15 minutes. It moves at 50 km/h whenever in motion. What distance is covered?

- A. 187.5 - CORRECT: Correct. Moving time= $4-15/60=3.75$ hours; distance= $50 \times 3.75=187.5$ km.
- B. 200 - INCORRECT: Incorrect: 200 does not follow the stated data. Moving time= $4-15/60=3.75$ hours; distance= $50 \times 3.75=187.5$ km.
- C. 137.5 - INCORRECT: Incorrect: 137.5 is an intermediate, sign or operation error. Moving time= $4-15/60=3.75$ hours; distance= $50 \times 3.75=187.5$ km.
- D. 237.5 - INCORRECT: Incorrect: 237.5 ignores a condition or uses the wrong base. Moving time= $4-15/60=3.75$ hours; distance= $50 \times 3.75=187.5$ km.

Method / explanation: Moving time= $4-15/60=3.75$ hours; distance= $50 \times 3.75=187.5$ km.

Trap: Remove halt time before multiplying by speed.

Q50. Answer: B | Speed and Distance

Over 260 km, how many hours are saved by increasing speed from 41 to 61 km/h?

- A. 13.0 - INCORRECT: Incorrect: 13.0 does not follow the stated data. Time saved = $260/41 - 260/61 = 2.08$ hours.
- B. 2.08 - CORRECT: Correct. Time saved = $260/41 - 260/61 = 2.08$ hours.
- C. 3.08 - INCORRECT: Incorrect: 3.08 is an intermediate, sign or operation error. Time saved = $260/41 - 260/61 = 2.08$ hours.
- D. 1.08 - INCORRECT: Incorrect: 1.08 ignores a condition or uses the wrong base. Time saved = $260/41 - 260/61 = 2.08$ hours.

Method / explanation: Time saved = $260/41 - 260/61 = 2.08$ hours.

Trap: Subtract travel times, not speeds.

Q51. Answer: C | Speed and Distance

A 190-metre train moves at 54 km/h. How many seconds does it take to clear a 95-metre platform?

- A. 12.67 - INCORRECT: Incorrect: 12.67 does not follow the stated data. Total distance= $190+95$; $54 \text{ km/h}=15 \text{ m/s}$; time= 19.00 s .
- B. 21.0 - INCORRECT: Incorrect: 21.0 is an intermediate, sign or operation error. Total distance= $190+95$; $54 \text{ km/h}=15 \text{ m/s}$; time= 19.00 s .
- C. 19.0 - CORRECT: Correct. Total distance= $190+95$; $54 \text{ km/h}=15 \text{ m/s}$; time= 19.00 s .
- D. 17.0 - INCORRECT: Incorrect: 17.0 ignores a condition or uses the wrong base. Total distance= $190+95$; $54 \text{ km/h}=15 \text{ m/s}$; time= 19.00 s .

Method / explanation: Total distance= $190+95$; $54 \text{ km/h}=15 \text{ m/s}$; time= 19.00 s .

Trap: To clear a platform, use train length plus platform length.

Q52. Answer: D | Speed and Distance

A boat's still-water speed is 11 km/h and stream speed is 3 km/h. How long does it take to travel 16 km upstream?

A. 1.1428571428571428 - INCORRECT: Incorrect: 1.1428571428571428 does not follow the stated data. Upstream speed= $11-3=8$ km/h; time= $16/8=2$ hours.

B. 8 - INCORRECT: Incorrect: 8 is an intermediate, sign or operation error. Upstream speed= $11-3=8$ km/h; time= $16/8=2$ hours.

C. 1.4545454545454546 - INCORRECT: Incorrect: 1.4545454545454546 ignores a condition or uses the wrong base. Upstream speed= $11-3=8$ km/h; time= $16/8=2$ hours.

D. 2 - CORRECT: Correct. Upstream speed= $11-3=8$ km/h; time= $16/8=2$ hours.

Method / explanation: Upstream speed= $11-3=8$ km/h; time= $16/8=2$ hours.

Trap: First obtain upstream speed, then divide distance by that speed.

Q53. Answer: A | Speed and Distance

A runner completes 4 laps of a square field of side 21 m in 8 minutes. What is the runner's speed in metres per minute?

A. 42.0 - CORRECT: Correct. Total distance= $4 \times 4 \times 21$; divide by 8 minutes to get 42.0 m/min.

B. 84 - INCORRECT: Incorrect: 84 does not follow the stated data. Total distance= $4 \times 4 \times 21$; divide by 8 minutes to get 42.0 m/min.

C. 55.125 - INCORRECT: Incorrect: 55.125 is an intermediate, sign or operation error. Total distance= $4 \times 4 \times 21$; divide by 8 minutes to get 42.0 m/min.

D. 21.0 - INCORRECT: Incorrect: 21.0 ignores a condition or uses the wrong base. Total distance= $4 \times 4 \times 21$; divide by 8 minutes to get 42.0 m/min.

Method / explanation: Total distance= $4 \times 4 \times 21$; divide by 8 minutes to get 42.0 m/min.

Trap: Compute total distance and total time before cancelling common factors.

Q54. Answer: B | Speed and Distance

A car covers 120 km at 41 km/h and 180 km at 61 km/h. What is average speed?

A. 51.0 - INCORRECT: Incorrect: 51.0 does not follow the stated data. Average speed = total distance/total time = $300/(120/41+180/61) = 51.04$.

B. 51.04 - CORRECT: Correct. Average speed = total distance/total time = $300/(120/41+180/61) = 51.04$.

C. 53.04 - INCORRECT: Incorrect: 53.04 is an intermediate, sign or operation error. Average speed = total distance/total time = $300/(120/41+180/61) = 51.04$.

D. 49.04 - INCORRECT: Incorrect: 49.04 ignores a condition or uses the wrong base. Average speed = total distance/total time = $300/(120/41+180/61) = 51.04$.

Method / explanation: Average speed = total distance/total time = $300/(120/41+180/61) = 51.04$.

Trap: Speeds need time weights.

Q55. Answer: C | Data Interpretation

A table reports district outputs (A=85, B=99, C=113, D=127). What is the total?

A. 414 - INCORRECT: Incorrect: 414 does not follow the stated data. Add all four entries: $85+99+113+127=424$.

B. 434 - INCORRECT: Incorrect: 434 is an intermediate, sign or operation error. Add all four entries: $85+99+113+127=424$.

C. 424 - CORRECT: Correct. Add all four entries: $85+99+113+127=424$.

D. 127 - INCORRECT: Incorrect: 127 ignores a condition or uses the wrong base. Add all four entries: $85+99+113+127=424$.

Method / explanation: Add all four entries: $85+99+113+127=424$.

Trap: Use every row exactly once.

Q56. Answer: D | Data Interpretation

Using the table (A=85, B=99, C=113, D=127), what is the difference between the highest and lowest values?

A. 127 - INCORRECT: Incorrect: 127 does not follow the stated data. Range = $127-85=42$.

B. 85 - INCORRECT: Incorrect: 85 is an intermediate, sign or operation error. Range = $127-85=42$.

C. 212 - INCORRECT: Incorrect: 212 ignores a condition or uses the wrong base. Range = $127-85=42$.

D. 42 - CORRECT: Correct. Range = $127-85=42$.

Method / explanation: Range = $127-85=42$.

Trap: Range is a difference, not an endpoint.

Q57. Answer: A | Data Interpretation

Using the table (A=85, B=99, C=113, D=127), what is the average of the four values?

A. 106.0 - CORRECT: Correct. Average = $424/4=106.0$.

B. 424 - INCORRECT: Incorrect: 424 does not follow the stated data. Average = $424/4=106.0$.

C. 212.0 - INCORRECT: Incorrect: 212.0 is an intermediate, sign or operation error. Average = $424/4=106.0$.

D. 42 - INCORRECT: Incorrect: 42 ignores a condition or uses the wrong base. Average = $424/4=106.0$.

Method / explanation: Average = $424/4=106.0$.

Trap: Divide the total by four.

Q58. Answer: B | Data Interpretation

Using the table (A=85, B=99, C=113, D=127), by what percentage does D exceed A?

A. 33.07 - INCORRECT: Incorrect: 33.07 does not follow the stated data. Percentage increase = $(127-85)/85 \times 100$.

B. 49.41 - CORRECT: Correct. Percentage increase = $(127-85)/85 \times 100$.

C. 42 - INCORRECT: Incorrect: 42 is an intermediate, sign or operation error. Percentage increase = $(127-85)/85 \times 100$.

D. 149.41 - INCORRECT: Incorrect: 149.41 ignores a condition or uses the wrong base. Percentage increase = $(127-85)/85 \times 100$.

Method / explanation: Percentage increase = $(127-85)/85 \times 100$.

Trap: Use the original value A as denominator.

Q59. Answer: C | Data Interpretation

Year 1 values are A=85, B=99, C=113, D=127; Year 2 values are A=106, B=115, C=124, D=133. Which district has the largest absolute increase?

A. B - INCORRECT: Incorrect: B does not follow the stated data. The increases are 21, 16, 11 and 6; A is largest.

B. C - INCORRECT: Incorrect: C is an intermediate, sign or operation error. The increases are 21, 16, 11 and 6; A is largest.

C. A - CORRECT: Correct. The increases are 21, 16, 11 and 6; A is largest.

D. D - INCORRECT: Incorrect: D ignores a condition or uses the wrong base. The increases are 21, 16, 11 and 6; A is largest.

Method / explanation: The increases are 21, 16, 11 and 6; A is largest.

Trap: Check absolute increases before comparing totals.

Q60. Answer: D | Data Interpretation

Using Year 2 (A=106, B=115, C=124, D=133), what fraction of the total belongs to B, rounded to two decimals?

A. 0.27 - INCORRECT: Incorrect: 0.27 does not follow the stated data. B's share = $115/478 = 0.24$.

B. 4.16 - INCORRECT: Incorrect: 4.16 is an intermediate, sign or operation error. B's share = $115/478 = 0.24$.

C. 0.22 - INCORRECT: Incorrect: 0.22 ignores a condition or uses the wrong base. B's share = $115/478 = 0.24$.

D. 0.24 - CORRECT: Correct. B's share = $115/478 = 0.24$.

Method / explanation: B's share = $115/478 = 0.24$.

Trap: The denominator is the same-year total.

Q61. Answer: A | Data Interpretation

Using Year 1 (A=85, B=99, C=113, D=127), if each unit earns Rs 3, what is D's revenue?

A. 381 - CORRECT: Correct. Revenue = D units $\times$ rate = 127×3 .

- B. 130 - INCORRECT: Incorrect: 130 does not follow the stated data. Revenue = D units x rate = 127×3 .
- C. 127 - INCORRECT: Incorrect: 127 is an intermediate, sign or operation error. Revenue = D units x rate = 127×3 .
- D. 1272 - INCORRECT: Incorrect: 1272 ignores a condition or uses the wrong base. Revenue = D units x rate = 127×3 .

Method / explanation: Revenue = D units x rate = 127×3 .

Trap: Use D's value, not total output.

Q62. Answer: B | Data Interpretation

Using Year 1 (A=85, B=99, C=113, D=127), what is the ratio A:D in simplest form?

- A. 127:85 - INCORRECT: Incorrect: 127:85 does not follow the stated data. Divide 85:127 by gcd 1.
- B. 85:127 - CORRECT: Correct. Divide 85:127 by gcd 1.
- C. 170:254 - INCORRECT: Incorrect: 170:254 is an intermediate, sign or operation error. Divide 85:127 by gcd 1.
- D. 212:1 - INCORRECT: Incorrect: 212:1 ignores a condition or uses the wrong base. Divide 85:127 by gcd 1.

Method / explanation: Divide 85:127 by gcd 1.

Trap: Preserve the requested A-to-D order.

Q63. Answer: C | Logical Reasoning

All auditors are trained. No person lacking that status may certify. Which conclusion necessarily follows?

- A. Every trained person is one of the auditors. - INCORRECT: Incorrect: Every trained person is one of the auditors. does not follow the stated data. All auditors possess the stated status, so none can simultaneously lack it.
- B. Some auditors lack that status. - INCORRECT: Incorrect: Some auditors lack that status. is an intermediate, sign or operation error. All auditors possess the stated status, so none can simultaneously lack it.
- C. Every auditor is outside the class lacking that status. - CORRECT: Correct. All auditors possess the stated status, so none can simultaneously lack it.
- D. No trained person may certify. - INCORRECT: Incorrect: No trained person may certify. ignores a condition or uses the wrong base. All auditors possess the stated status, so none can simultaneously lack it.

Method / explanation: All auditors possess the stated status, so none can simultaneously lack it.

Trap: Do not reverse a universal proposition.

Q64. Answer: D | Logical Reasoning

If each letter is shifted forward by 1, how is INDIA coded?

- A. INDIA - INCORRECT: Incorrect: INDIA does not follow the stated data. Shift each letter independently by 1: INDIA -> JOEJB.
- B. BJEOJ - INCORRECT: Incorrect: BJEOJ is an intermediate, sign or operation error. Shift each letter independently by 1: INDIA -> JOEJB.
- C. JODJB - INCORRECT: Incorrect: JODJB ignores a condition or uses the wrong base. Shift each letter independently by 1: INDIA -> JOEJB.
- D. JOEJB - CORRECT: Correct. Shift each letter independently by 1: INDIA -> JOEJB.

Method / explanation: Shift each letter independently by 1: INDIA -> JOEJB.

Trap: Do not reverse the word.

Q65. Answer: A | Logical Reasoning

Complete the series: 3, 5, 9, 15, ?

- A. 23 - CORRECT: Correct. Successive increments are 2, 4, 6 and 8; next term is 23.
- B. 22 - INCORRECT: Incorrect: 22 does not follow the stated data. Successive increments are 2, 4, 6 and 8; next term is 23.
- C. 24 - INCORRECT: Incorrect: 24 is an intermediate, sign or operation error. Successive increments are 2, 4, 6 and 8; next term is 23.
- D. 21 - INCORRECT: Incorrect: 21 ignores a condition or uses the wrong base. Successive increments are 2, 4, 6 and 8; next term is 23.

Method / explanation: Successive increments are 2, 4, 6 and 8; next term is 23.

Trap: Track differences, not ratios.

Q66. Answer: B | Logical Reasoning

Mira is the child of Arun's only son. How is Mira related to Arun?

- A. Child - INCORRECT: Incorrect: Child does not follow the stated data. The only son is the parent, so Mira is the elder's grandchild.
- B. Grandchild - CORRECT: Correct. The only son is the parent, so Mira is the elder's grandchild.
- C. Niece or nephew - INCORRECT: Incorrect: Niece or nephew is an intermediate, sign or operation error. The only son is the parent, so Mira is the elder's grandchild.
- D. Sibling - INCORRECT: Incorrect: Sibling ignores a condition or uses the wrong base. The only son is the parent, so Mira is the elder's grandchild.

Method / explanation: The only son is the parent, so Mira is the elder's grandchild.

Trap: Resolve one relationship at a time.

Q67. Answer: C | Logical Reasoning

Some books are reports. All reports are archived. Which conclusion follows?

- A. All books are archived - INCORRECT: Incorrect: All books are archived does not follow the stated data. The members of books that are reports inherit the property archived.
- B. No books are archived - INCORRECT: Incorrect: No books are archived is an intermediate, sign or operation error. The members of books that are reports inherit the property archived.
- C. Some books are archived - CORRECT: Correct. The members of books that are reports inherit the property archived.
- D. Some archived are not reports - INCORRECT: Incorrect: Some archived are not reports ignores a condition or uses the wrong base. The members of books that are reports inherit the property archived.

Method / explanation: The members of books that are reports inherit the property archived.

Trap: Do not universalise a particular premise.

Q68. Answer: D | Logical Reasoning

In a code, GOVERN is written as HPWFSO by shifting each letter forward once. How is STATE written?

- A. STATE - INCORRECT: Incorrect: STATE does not follow the stated data. Shift every letter of STATE forward once to obtain TUBUF.
- B. FUBUT - INCORRECT: Incorrect: FUBUT is an intermediate, sign or operation error. Shift every letter of STATE forward once to obtain TUBUF.
- C. AAAAA - INCORRECT: Incorrect: AAAAA ignores a condition or uses the wrong base. Shift every letter of STATE forward once to obtain TUBUF.
- D. TUBUF - CORRECT: Correct. Shift every letter of STATE forward once to obtain TUBUF.

Method / explanation: Shift every letter of STATE forward once to obtain TUBUF.

Trap: Apply the same shift to every letter.

Q69. Answer: A | Arrangements

A, B, C and D sit in a row. A is left of B; C is right of B; D is left of A. Who is second from the left?

- A. A - CORRECT: Correct. The only order consistent with all relations is D-A-B-C.
- B. B - INCORRECT: Incorrect: B does not follow the stated data. The only order consistent with all relations is D-A-B-C.
- C. C - INCORRECT: Incorrect: C is an intermediate, sign or operation error. The only order consistent with all relations is D-A-B-C.
- D. D - INCORRECT: Incorrect: D ignores a condition or uses the wrong base. The only order consistent with all relations is D-A-B-C.

Method / explanation: The only order consistent with all relations is D-A-B-C.

Trap: Build the full order before selecting a position.

Q70. Answer: B | Arrangements

Four tasks P, Q, R and S occur consecutively. P precedes Q; R follows Q; S precedes P. Which task is third?

- A. P - INCORRECT: Incorrect: P does not follow the stated data. The forced order is S-P-Q-R, so Q is third.
- B. Q - CORRECT: Correct. The forced order is S-P-Q-R, so Q is third.
- C. R - INCORRECT: Incorrect: R is an intermediate, sign or operation error. The forced order is S-P-Q-R, so Q is third.
- D. S - INCORRECT: Incorrect: S ignores a condition or uses the wrong base. The forced order is S-P-Q-R, so Q is third.

Method / explanation: The forced order is S-P-Q-R, so Q is third.

Trap: Precedes means earlier, not immediately earlier unless stated.

Q71. Answer: C | Arrangements

Five persons stand by height. K is taller than L but shorter than M. N is shorter than L. O is taller than M. Who is tallest?

- A. M - INCORRECT: Incorrect: M does not follow the stated data. The chain is $O > M > K > L > N$.
- B. K - INCORRECT: Incorrect: K is an intermediate, sign or operation error. The chain is $O > M > K > L > N$.
- C. O - CORRECT: Correct. The chain is $O > M > K > L > N$.
- D. N - INCORRECT: Incorrect: N ignores a condition or uses the wrong base. The chain is $O > M > K > L > N$.

Method / explanation: The chain is $O > M > K > L > N$.

Trap: Combine all comparisons into one chain.

Q72. Answer: D | Arrangements

Review is after Finance but before Health. Education is before Finance. Which is earliest?

- A. Finance - INCORRECT: Incorrect: Finance does not follow the stated data. The order is Education, Finance, Review, Health.
- B. Review - INCORRECT: Incorrect: Review is an intermediate, sign or operation error. The order is Education, Finance, Review, Health.
- C. Health - INCORRECT: Incorrect: Health ignores a condition or uses the wrong base. The order is Education, Finance, Review, Health.
- D. Education - CORRECT: Correct. The order is Education, Finance, Review, Health.

Method / explanation: The order is Education, Finance, Review, Health.

Trap: Translate every before/after relation consistently.

Q73. Answer: A | Directions

A person faces North and turns right 1 time(s), each by 90 degrees. Which direction is faced?

- A. East - CORRECT: Correct. Each right turn advances one step clockwise; 1 turn(s) gives East.
- B. North - INCORRECT: Incorrect: North does not follow the stated data. Each right turn advances one step clockwise; 1 turn(s) gives East.
- C. South - INCORRECT: Incorrect: South is an intermediate, sign or operation error. Each right turn advances one step clockwise; 1 turn(s) gives East.
- D. West - INCORRECT: Incorrect: West ignores a condition or uses the wrong base. Each right turn advances one step clockwise; 1 turn(s) gives East.

Method / explanation: Each right turn advances one step clockwise; 1 turn(s) gives East.

Trap: Reduce turns modulo four.

Q74. Answer: B | Directions

A walker goes 3 km north and 4 km east. How far is the endpoint from the start?

- A. 7 km - INCORRECT: Incorrect: 7 km does not follow the stated data. Displacement = $\sqrt{3^2+4^2}=5$ km.
- B. 5 km - CORRECT: Correct. Displacement = $\sqrt{3^2+4^2}=5$ km.
- C. 1 km - INCORRECT: Incorrect: 1 km is an intermediate, sign or operation error. Displacement = $\sqrt{3^2+4^2}=5$ km.
- D. 12 km - INCORRECT: Incorrect: 12 km ignores a condition or uses the wrong base. Displacement = $\sqrt{3^2+4^2}=5$ km.

Method / explanation: Displacement = $\sqrt{3^2+4^2}=5$ km.

Trap: Distance walked and displacement differ.

Q75. Answer: C | Directions

A person starts facing East, turns left, then turns left again. Which direction is now faced?

- A. North - INCORRECT: Incorrect: North does not follow the stated data. Two left turns reverse the starting direction, giving West.
- B. East - INCORRECT: Incorrect: East is an intermediate, sign or operation error. Two left turns reverse the starting direction, giving West.
- C. West - CORRECT: Correct. Two left turns reverse the starting direction, giving West.
- D. South - INCORRECT: Incorrect: South ignores a condition or uses the wrong base. Two left turns reverse the starting direction, giving West.

Method / explanation: Two left turns reverse the starting direction, giving West.

Trap: Track orientation after each turn.

Q76. Answer: D | Directions

A point B is 5 km south of A, and C is 5 km east of B. In which direction is C from A?

- A. North-East - INCORRECT: Incorrect: North-East does not follow the stated data. Relative to A, C has positive east and negative north coordinates: south-east.
- B. South-West - INCORRECT: Incorrect: South-West is an intermediate, sign or operation error. Relative to A, C has positive east and negative north coordinates: south-east.
- C. North-West - INCORRECT: Incorrect: North-West ignores a condition or uses the wrong base. Relative to A, C has positive east and negative north coordinates: south-east.
- D. South-East - CORRECT: Correct. Relative to A, C has positive east and negative north coordinates: south-east.

Method / explanation: Relative to A, C has positive east and negative north coordinates: south-east.

Trap: Use a simple coordinate sketch.

Q77. Answer: A | Data Sufficiency

What is x ? Statement 1: $x+3=10$. Statement 2: $2x=14$.

- A. Each statement alone is sufficient. - CORRECT: Correct. Statement 1 gives $x=7$; statement 2 also gives $x=7$. Each alone is sufficient.
- B. Statement 1 alone is sufficient, but statement 2 alone is not. - INCORRECT: Incorrect: Statement 1 alone is sufficient, but statement 2 alone is not. does not follow the stated data. Statement 1 gives $x=7$; statement 2 also gives $x=7$. Each alone is sufficient.
- C. Statement 2 alone is sufficient, but statement 1 alone is not. - INCORRECT: Incorrect: Statement 2 alone is sufficient, but statement 1 alone is not. is an intermediate, sign or operation error. Statement 1 gives $x=7$; statement 2 also gives $x=7$. Each alone is sufficient.
- D. Both statements together are sufficient, but neither alone is sufficient. - INCORRECT: Incorrect: Both statements together are sufficient, but neither alone is sufficient. ignores a condition or uses the wrong base. Statement 1 gives $x=7$; statement 2 also gives $x=7$. Each alone is sufficient.

Method / explanation: Statement 1 gives $x=7$; statement 2 also gives $x=7$. Each alone is sufficient.

Trap: Test each statement separately before combining.

Q78. Answer: B | Data Sufficiency

What is the area of a rectangle? Statement 1: its length is 8 cm. Statement 2: its breadth is 5 cm.

- A. Statement 1 alone is sufficient, but statement 2 alone is not. - INCORRECT: Incorrect: Statement 1 alone is sufficient, but statement 2 alone is not. does not follow the stated data. Area requires both length and breadth; neither statement alone is enough.
- B. Both statements together are sufficient, but neither alone is sufficient. - CORRECT: Correct. Area requires both length and breadth; neither statement alone is enough.
- C. Statement 2 alone is sufficient, but statement 1 alone is not. - INCORRECT: Incorrect: Statement 2 alone is sufficient, but statement 1 alone is not. is an intermediate, sign or operation error. Area requires both length and breadth; neither statement alone is enough.

D. Each statement alone is sufficient. - INCORRECT: Incorrect: Each statement alone is sufficient. ignores a condition or uses the wrong base. Area requires both length and breadth; neither statement alone is enough.

Method / explanation: Area requires both length and breadth; neither statement alone is enough.

Trap: Do not import an unstated shape relation.

Q79. Answer: C | Data Sufficiency

Is integer n even? Statement 1: n is divisible by 8. Statement 2: n is divisible by 4.

A. Statement 1 alone is sufficient, but statement 2 alone is not. - INCORRECT: Incorrect: Statement 1 alone is sufficient, but statement 2 alone is not. does not follow the stated data. Either divisibility condition independently implies n is even.

B. Statement 2 alone is sufficient, but statement 1 alone is not. - INCORRECT: Incorrect: Statement 2 alone is sufficient, but statement 1 alone is not. is an intermediate, sign or operation error. Either divisibility condition independently implies n is even.

C. Each statement alone is sufficient. - CORRECT: Correct. Either divisibility condition independently implies n is even.

D. Both statements together are sufficient, but neither alone is sufficient. - INCORRECT: Incorrect: Both statements together are sufficient, but neither alone is sufficient. ignores a condition or uses the wrong base. Either divisibility condition independently implies n is even.

Method / explanation: Either divisibility condition independently implies n is even.

Trap: Sufficiency does not require both statements to be equally strong.

Q80. Answer: D | Data Sufficiency

What is y ? Statement 1: y is positive. Statement 2: $y^2=49$.

A. Statement 1 alone is sufficient, but statement 2 alone is not. - INCORRECT: Incorrect: Statement 1 alone is sufficient, but statement 2 alone is not. does not follow the stated data. Statement 2 gives $y=\text{plus or minus } 7$; statement 1 selects $y=7$. Together they are sufficient.

B. Statement 2 alone is sufficient, but statement 1 alone is not. - INCORRECT: Incorrect: Statement 2 alone is sufficient, but statement 1 alone is not. is an intermediate, sign or operation error. Statement 2 gives $y=\text{plus or minus } 7$; statement 1 selects $y=7$. Together they are sufficient.

C. Each statement alone is sufficient. - INCORRECT: Incorrect: Each statement alone is sufficient. ignores a condition or uses the wrong base. Statement 2 gives $y=\text{plus or minus } 7$; statement 1 selects $y=7$. Together they are sufficient.

D. Both statements together are sufficient, but neither alone is sufficient. - CORRECT: Correct. Statement 2 gives $y=\text{plus or minus } 7$; statement 1 selects $y=7$. Together they are sufficient.

Method / explanation: Statement 2 gives $y=\text{plus or minus } 7$; statement 1 selects $y=7$. Together they are sufficient.

Trap: A square equation has two real roots unless sign is fixed.

END OF DETAILED SOLUTIONS